Chapter 3

Rectifiers

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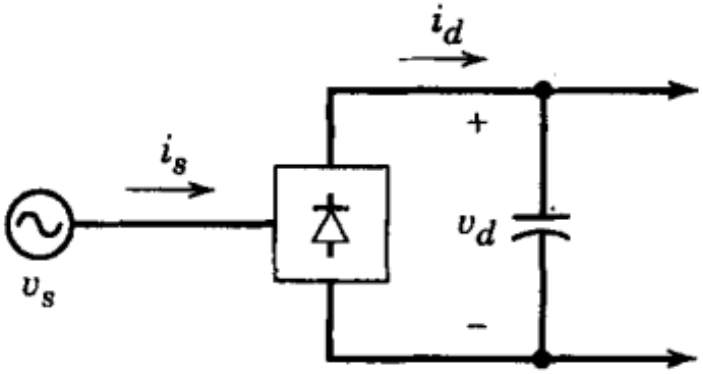
Line-frequency diode rectifier

Rectifier: An electrical device that converts alternating current (AC) to direct current (DC)

Line-frequency diode rectifier:

- Conversion: Line-frequency AC to uncontrolled DC
- Applications: power supplies, ac motor drive, variable speed DC drives, battery chargers,…
- Low cost, popular
- Large capacitor needed for ripple-free DC voltage
- Draw highly distorted current from utility

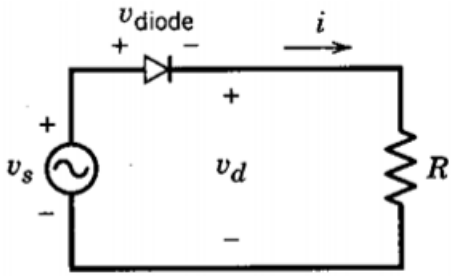
Note: Diodes are assumed to be ideal in the analysis of rectifiers



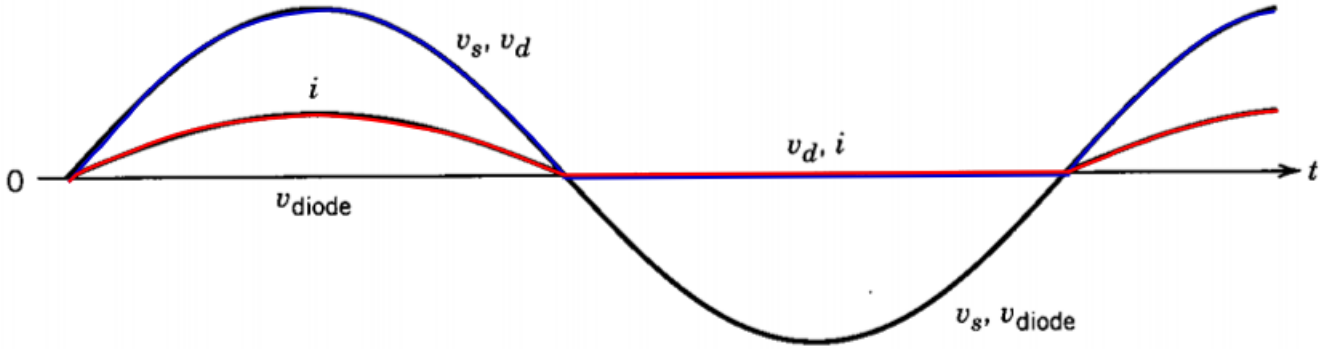
Single-phase diode rectifiers

Pure resitive load:

- Very simple, basic rectifier
- Diode conducts (forward-biased) for the positive half-cycle of the source
- Large ripple in v_d



(a)

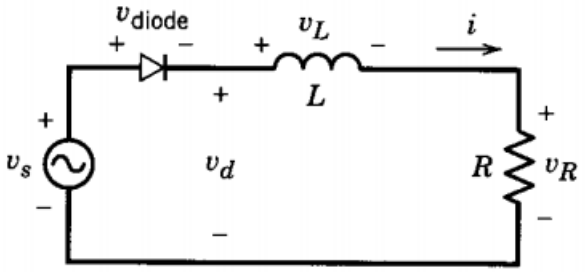


(b)

Single-phase diode rectifiers (half-wave rectifiers)

Inductive load:

- Industrial load typically contain inductance as well as resistance
- KVL when the diode is forward-biased

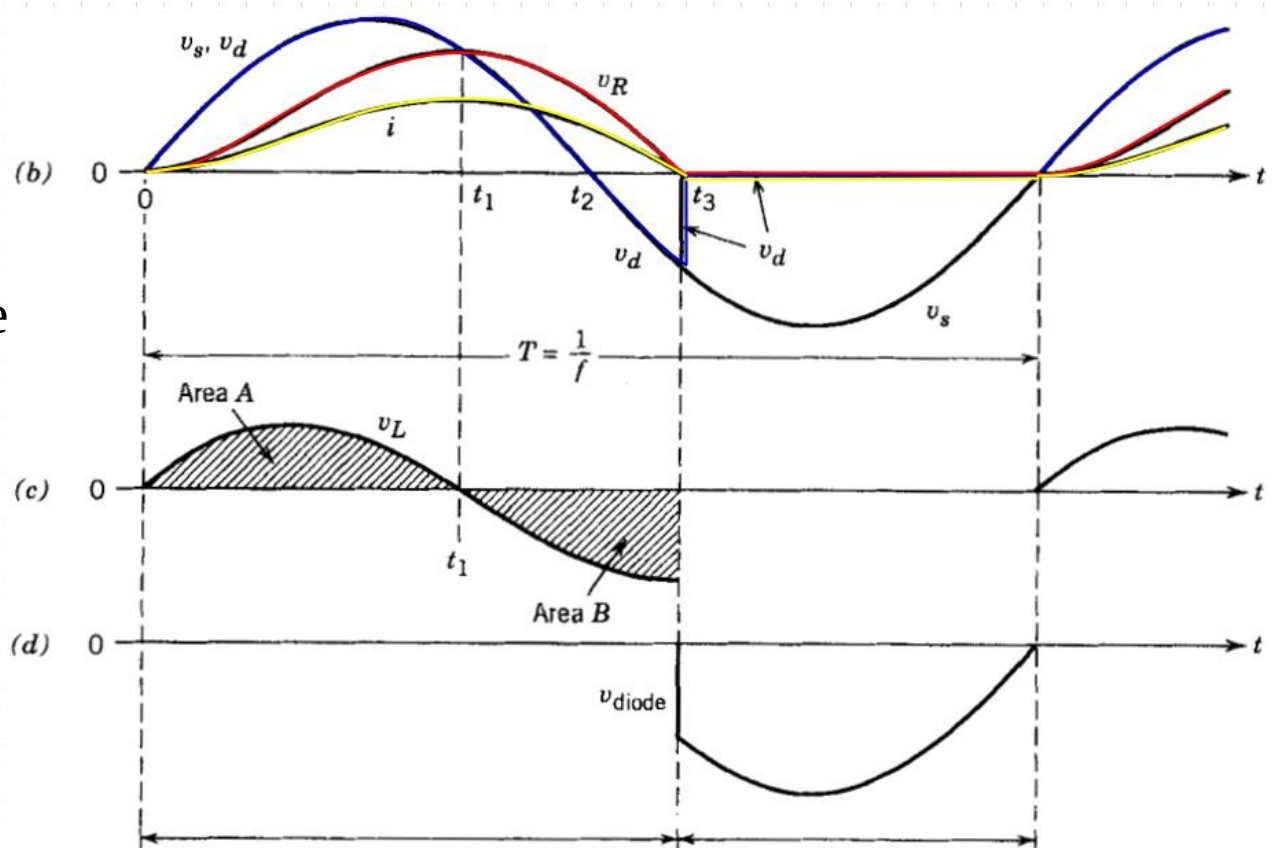


$$v_s = v_R + v_L$$

$$= Ri + L\frac{di}{dt}$$

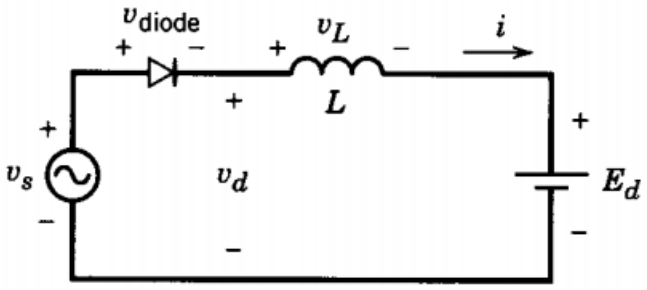
- Average voltage across L

$$V_{L,avr} = \frac{1}{T}\int_0^{t_3} v_L dt = 0$$



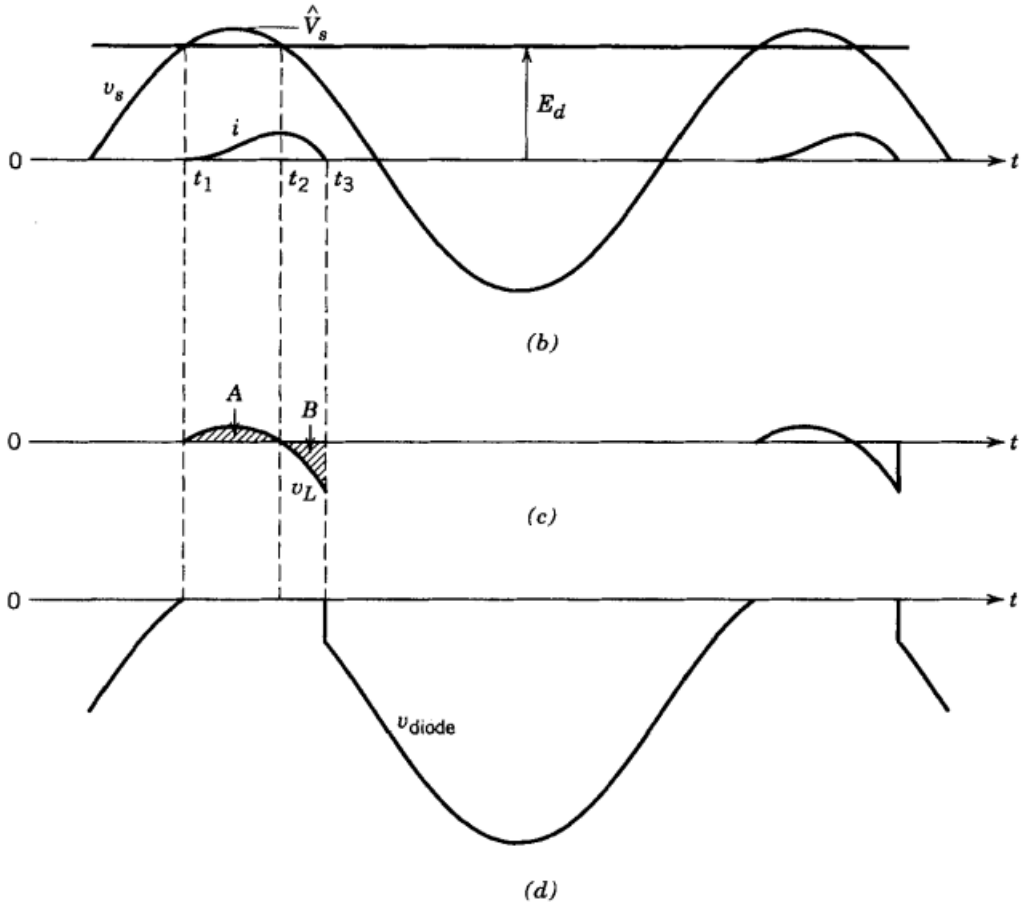
Single-phase diode rectifiers (half-wave rectifiers)

Load with internal DC voltage:



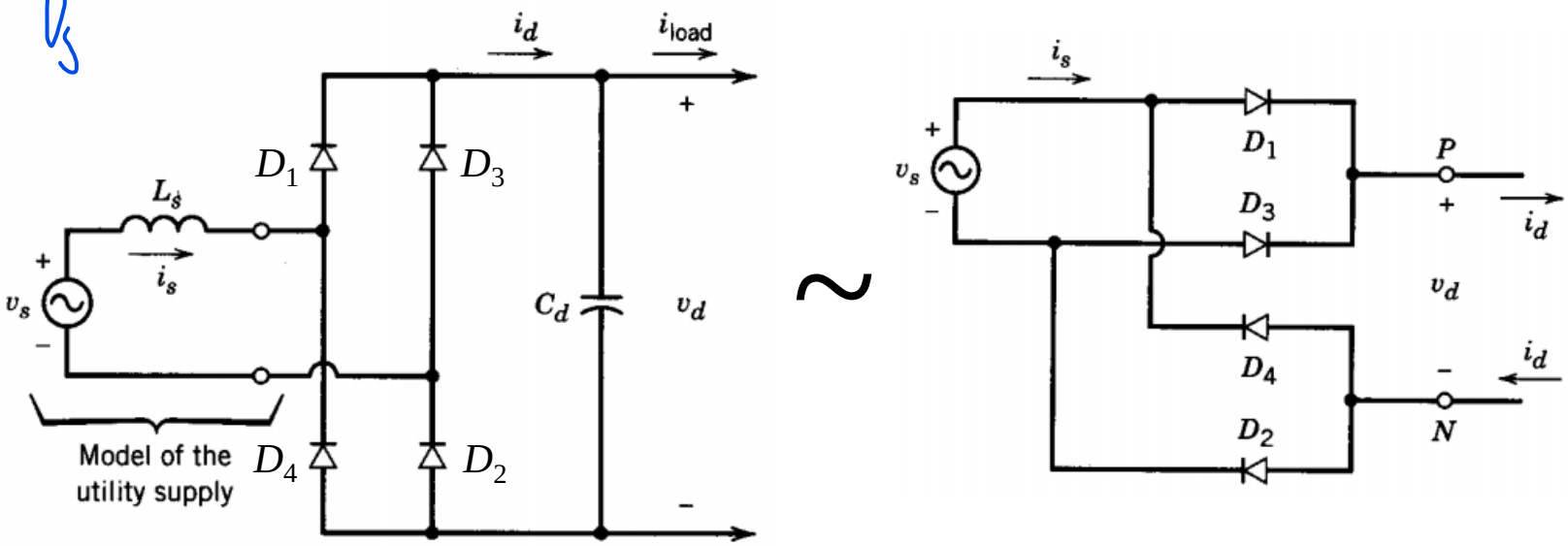
- The diode is forward-biased if $v_s > E_d$
- Average voltage across L

$$V_{L,avr} = \frac{1}{T}\int_{t_1}^{t_3} v_L dt = 0$$



Single-phase diode bridge rectifiers (full-wave rectifiers)

- Full-wave rectifiers are more commond to compared with half-wave rectifiers
- The utility supply is modeled as voltage source in series with internal impedance (primarily inductive)

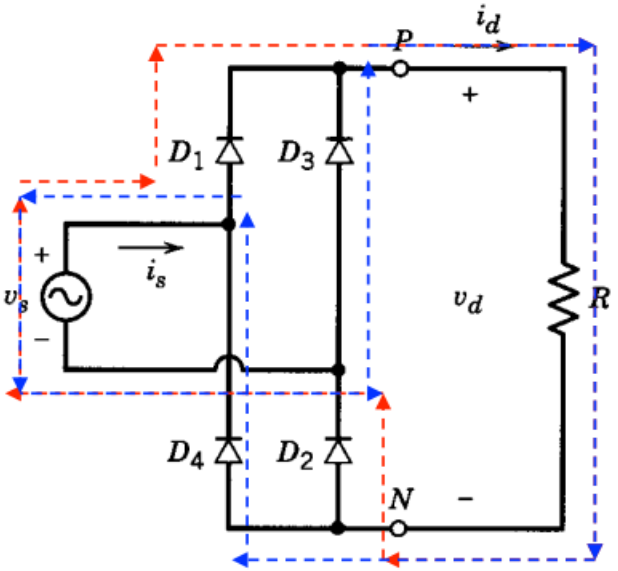


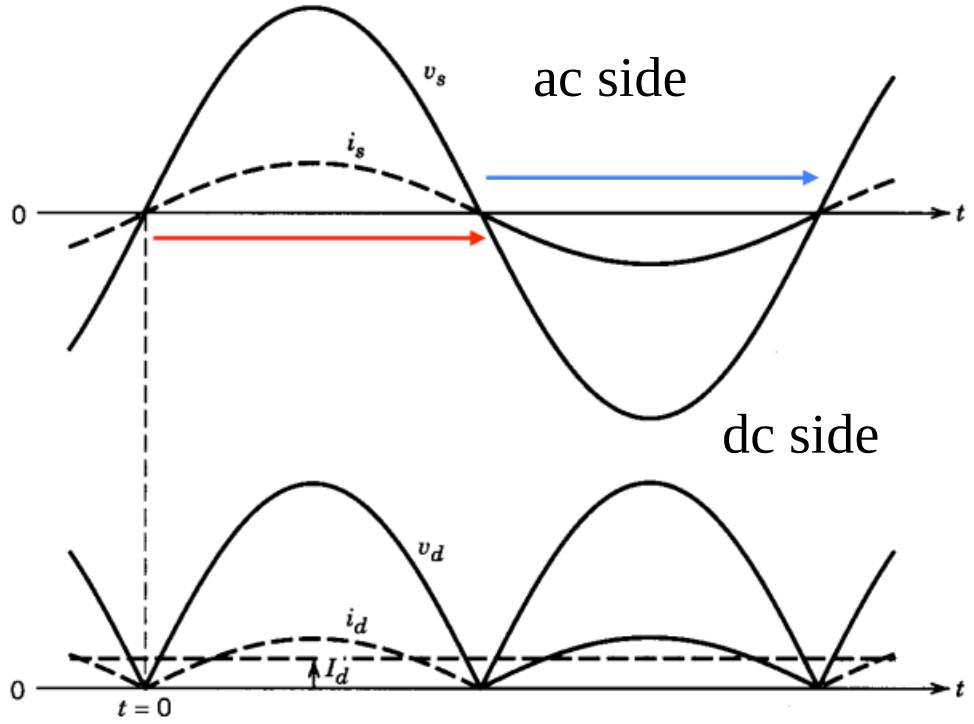
Conducting rules:

- Diode with highest anode potential is conducting
- Diode with lowest cathode potential is conducting

Single-phase diode bridge rectifiers (full-wave rectifiers)

Resistive load





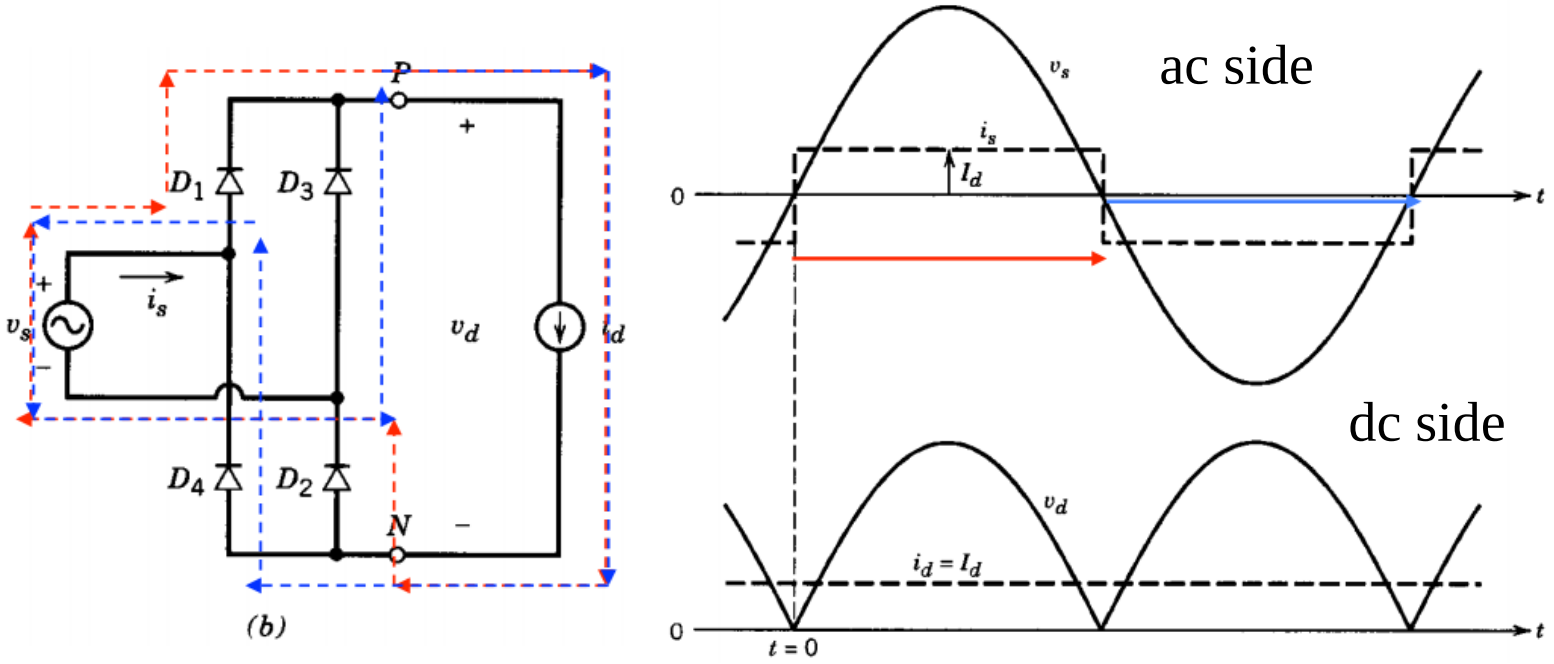
Average output voltage

$$V_{d0} = \frac{1}{T/2}\int_0^{T/2}\sqrt{2}V_s \sin\omega t dt = \frac{2}{\pi}\sqrt{2}V_s = 0.9V_s$$

Single-phase diode bridge rectifiers (full-wave rectifiers)

Inductive load (large inductor)

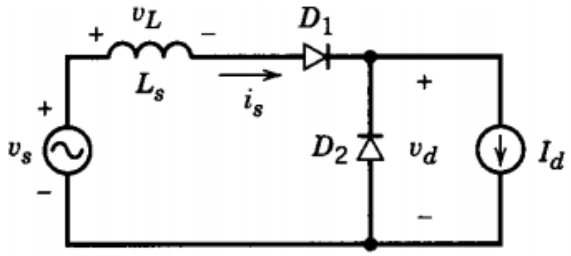
I_d is the approximation to a situation where the large inductor is connected in series at dc output of the rectifier for filtering

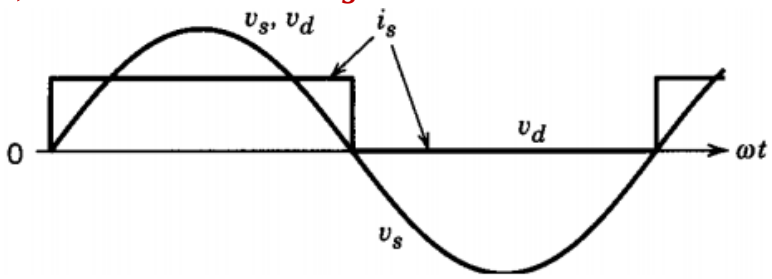


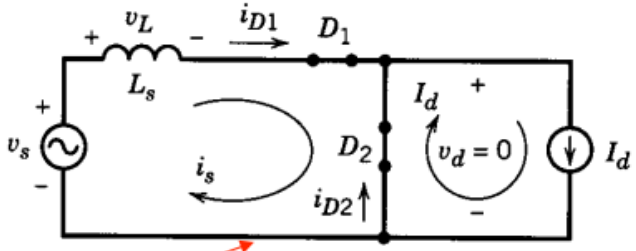
Note: if $L_s = 0$, is changes from I_d to $-I_d$ (and vice versa) instantaneously

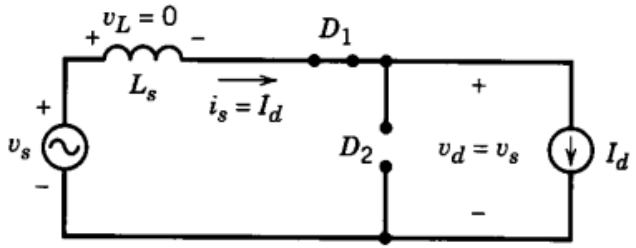
Single-phase diode bridge rectifiers (full-wave rectifiers)

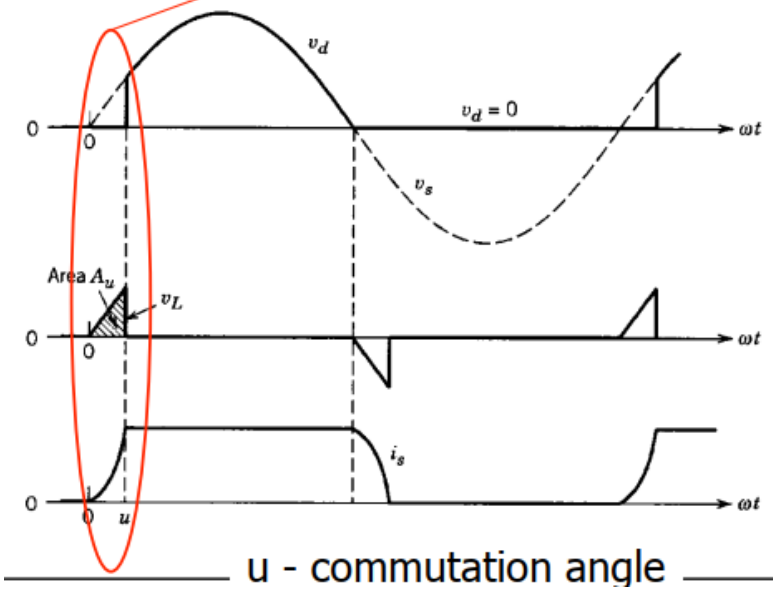
Inductive load (large inductor) – Effect of L_s











u - commutation angle

$$v_L = \sqrt{2}\ V_s\ \sin\omega t = L_s \frac{di_s}{dt} \qquad 0 < \omega t < u$$

$$i_s(t) = i_s(0) + \frac{1}{L_s}\int_0^t \sqrt{2}\ V_s\ \sin\omega t\ d(t)$$

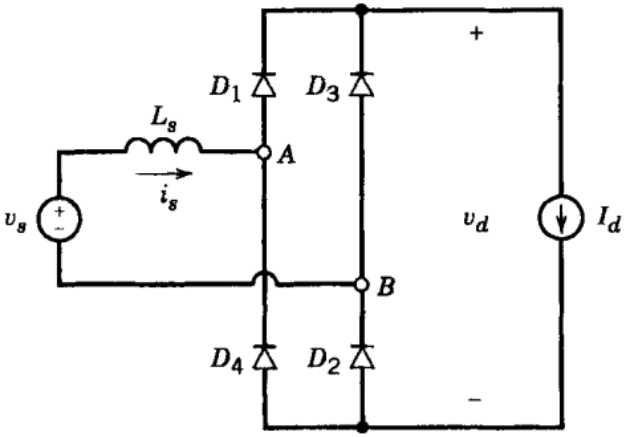
$$i_s(0) = 0 \qquad i_s(u/\omega) = I_d$$

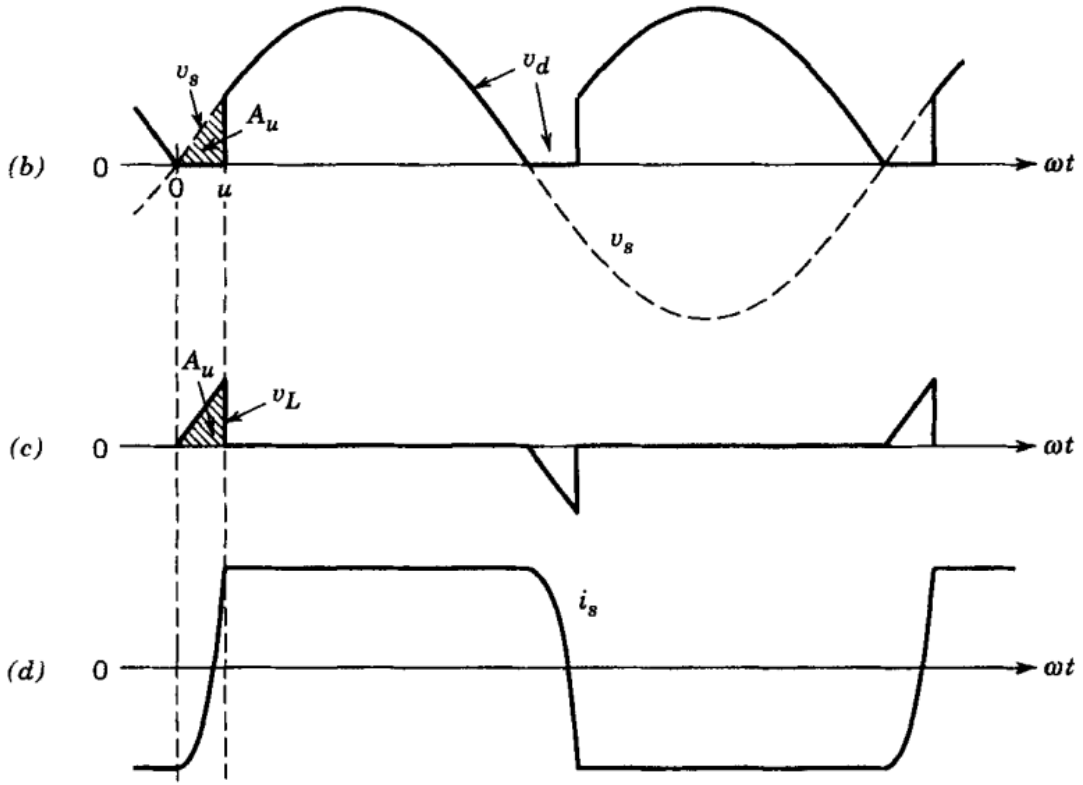
$$\omega L_s\ I_d = \sqrt{2}\ V_s\ (1 - \cos u)$$

$$\cos u = 1 - \frac{\omega L_s I_d}{\sqrt{2}\ V_s} \qquad A_u = \int_0^u \sqrt{2}\ V_s \sin\omega t\ d(\omega t)$$

Single-phase diode bridge rectifiers (full-wave rectifiers)

Inductive load (large inductor) – Effect of L_s

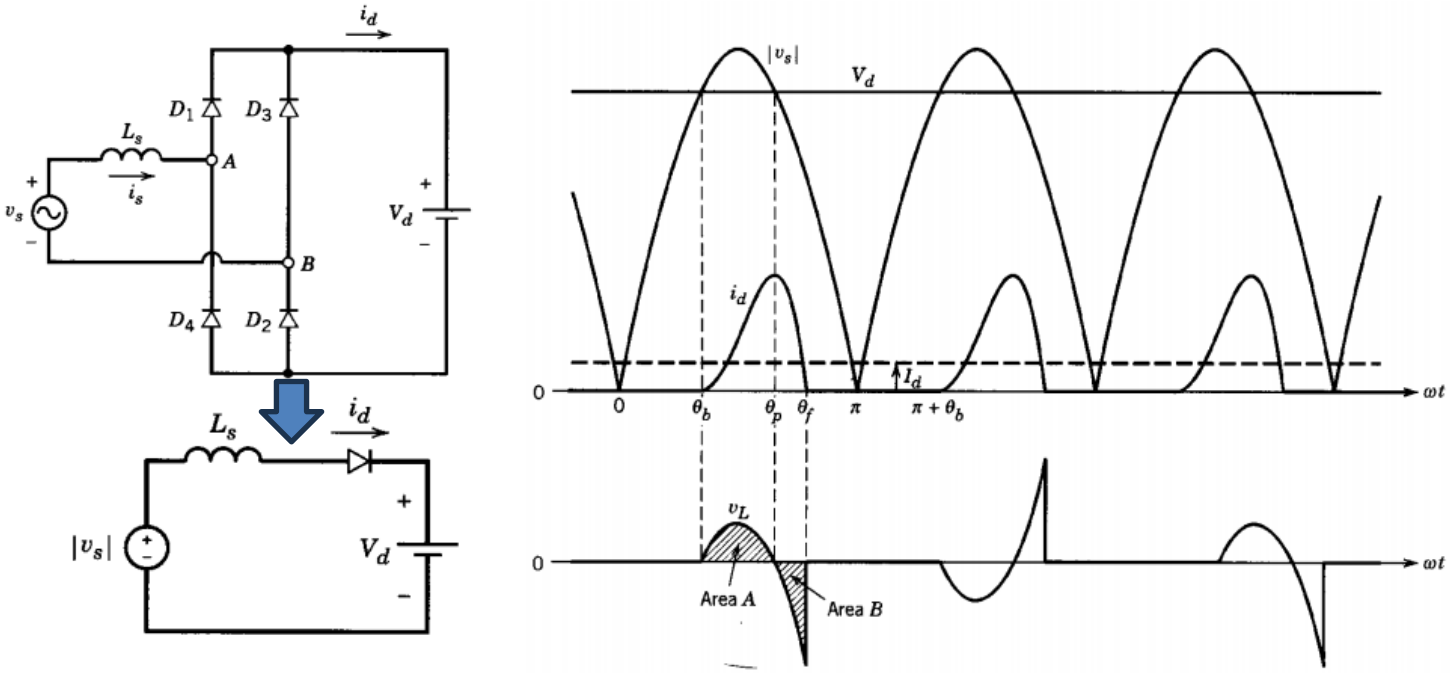




Single-phase diode bridge rectifiers (full-wave rectifiers)

Capacitive load (large capacitor)

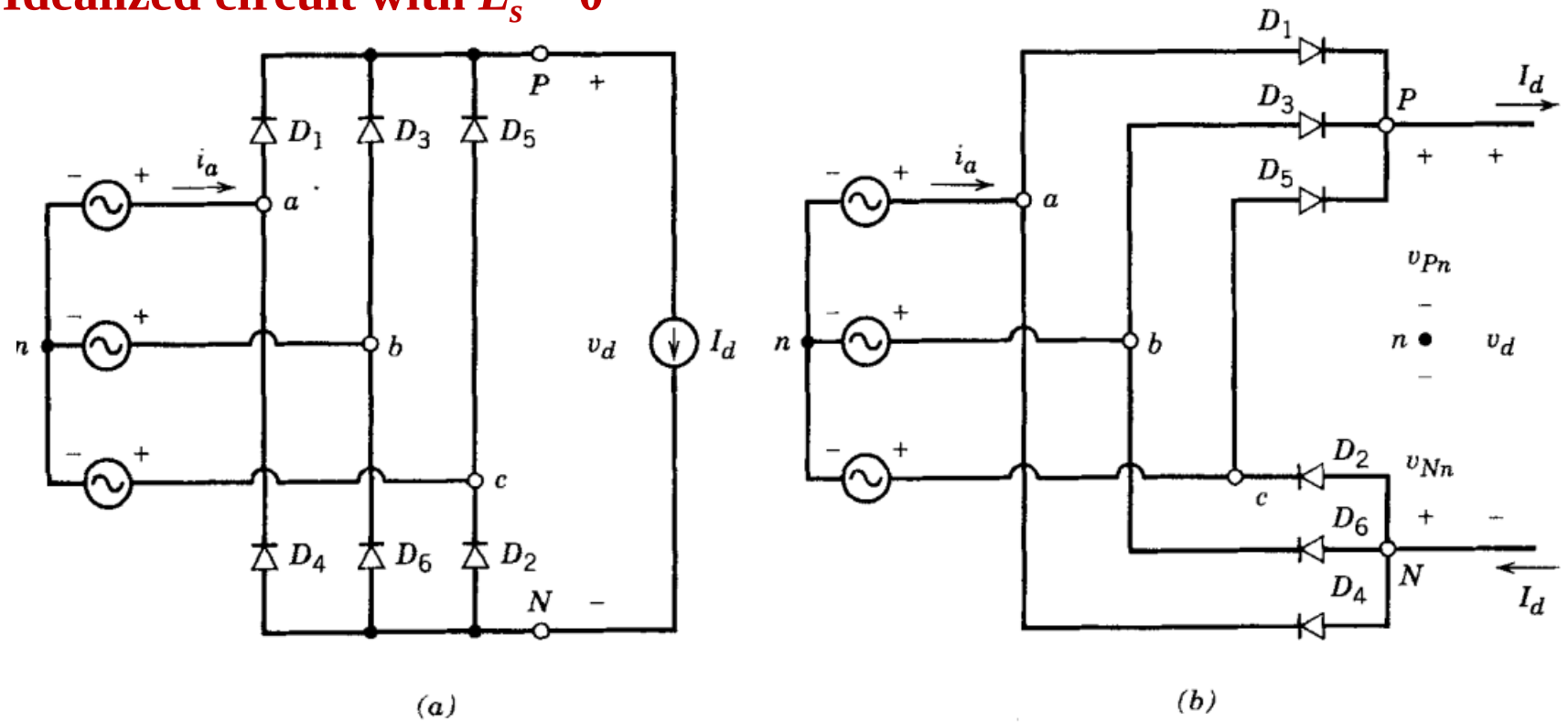
V_d is the approximation to a situation where the large capacitor is connected in series at dc output of the rectifier for filtering



- When v_s exceeds V_d at θ_b, D_1, D_2 start to conduct
- The current i_d reaches its peak at θ_p, beyond which v_L becomes negative
- The current becomes zero at θ_f when the volt-cross areas A and B become equal and negative of each other

Three-phase, full-bridge rectifiers

Idealized circuit with $L_s = 0$



Conducting rules:

- In the top group, the diode with its anode at the highest potential will conduct, the other two become reversed biased
- In the bottom group, the diode with its cathode at the lowest potential will conduct, the other two become reversed biased

Three-phase, full-bridge rectifiers

Idealized circuit with $L_s = 0$

$$v_d = v_{Pn} - v_{Nn}$$

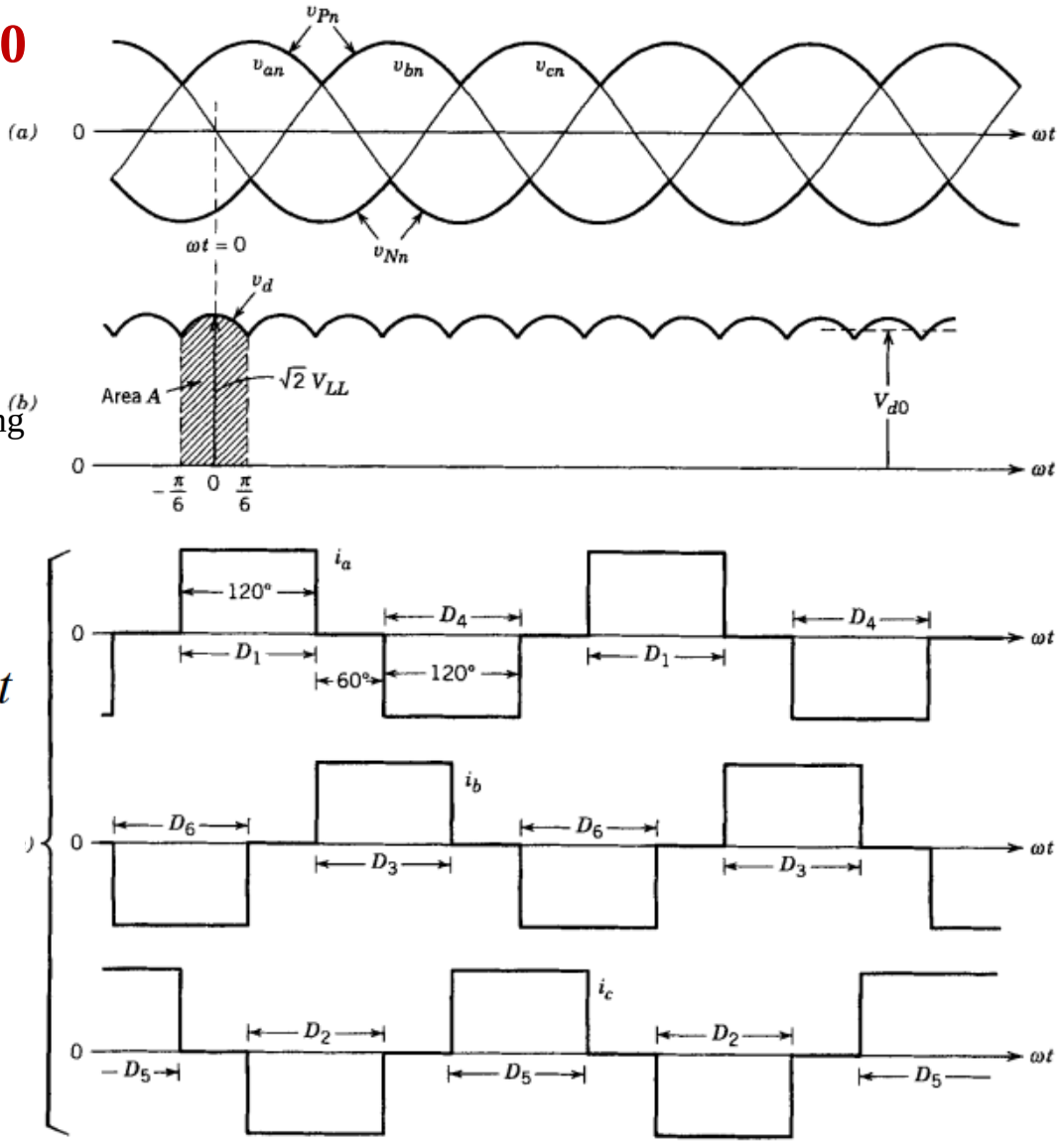
$$i_a = \begin{cases} I_d & \text{when D1 is conducting} \\ -I_d & \text{when D4 is conducting} \\ 0 & \text{when neither D1 or D4 is conducting} \end{cases}$$

For: $-\frac{1}{6}\pi < \omega t < \frac{1}{6}\pi$

$$v_d(t) = v_{ab}(t) = \sqrt{2}\ V_{LL}\ \cos \omega t$$

$$V_{do} = \frac{1}{\pi/3} \int_{-\pi/6}^{\pi/6} \sqrt{2}\, V_{LL} \cos \omega t$$

$$= \frac{3}{\pi} \sqrt{2}\, V_{LL} = 1.35\, V_{LL}$$



Three-phase, full-bridge rectifiers

Effect of L_s on current commutation

